

Mohamed Elgazar

Tanta, Egypt | [Gmail](#) | [Phone](#) | [Portfolio](#) | [LinkedIn](#) | [GitHub](#) | [Kaggle](#)

Summary

A highly motivated AI engineer with hands-on experience in machine learning, deep learning, and data-driven solutions. Skilled in developing end-to-end AI applications, from model training to deployment using modern tools and frameworks. Having experience in full-stack web development to build intuitive user interfaces and seamless integrations. Passionate about solving real-world problems through innovative AI approaches, with a strong foundation in Python, data analysis, and cloud deployment. Proven ability to work collaboratively in teams and deliver impactful, production-ready projects.

Education

Tanta University, BS in AI Engineering Oct 2022 – May 2027

- GPA: 3.96/4.0
- **Coursework:** Machine Learning, Deep Learning, Neural Networks, Data Structures, Operating Systems, Computer Architecture, Databases, Microprocessors & Microcontrollers.

Experience

Full-Time Developer at Meshkode – Remote Aug 2025 - Present

- Gained strong foundations in frontend development using HTML, CSS, JavaScript, and Angular, focusing on scalable and maintainable user interfaces.
- Applied modern frontend testing practices with Jest, React Testing Library (RTL), and Playwright to ensure high code quality.
- Developed hands-on experience in AI systems, including LangChain, LangGraph, prompt engineering, vector databases, and RAG.
- Worked with agent-based AI workflows and developer tools such as Agents.md and Continue Dev .
- Built and maintained development Environments using GitHub Actions, Temporal workflows, and monorepo architectures.
- Gained practical experience with graph databases using Neo4j and Cypher for modeling and querying relations.

Machine Learning Engineer (Depi Training Program), Microsoft – Remote Jun 2025 – Jan 2026
Under the supervision of EYouth Co. *AI & Data Science Track*

- Completed a hands-on training program focused on Microsoft's AI and Data Science tools and pipelines.
- Built and deployed ML models using supervised learning algorithms to solve classification problems.
- Gained practical experience with Azure ML, data preprocessing, and model evaluation techniques.

Projects

EduBot - The Smartest AI Assistant [Live Demo](#) | [Github Repo](#)

- Developed an Arabic-first AI tutoring system that converts Educational PDFs into an interactive learning assistant using Retrieval-Augmented Generation (RAG).
- Implemented both standard and agentic RAG pipelines with LangGraph, enabling multi-step reasoning, memory usage, and web search.
- Built a curriculum-aware vector retrieval system using Qdrant and multilingual embeddings, filtered by subject, grade, and term.
- Designed a student memory module to personalize explanations and track learning progress across conversations.
- Integrated Arabic voice capabilities with speech-to-text and text-to-speech for hands-free interaction.
- **Tools Used:** Python, Flask, LangChain, LangGraph, Qdrant, Hugging Face, Groq API, React, Express, Docker.

Brain Tumor Classification and Care

[Github Repo](#)

- Improved a CNN-based model to classify brain tumors from MRI scans with high accuracy.
- Compared performance of TensorFlow and PyTorch implementations on the same dataset.
- Integrated a care recommendation system and built an LLaMA 3-powered chatbot for post-diagnosis assistance.
- Designed a user-friendly interface to visualize predictions, confidence scores, and interact with the care assistant.
- **Tools Used:** Python, TensorFlow, PyTorch, LLaMA 3, HTML/CSS, Flask, Open WebUI.

SVM Binary Classification

[Github Repo](#)

- Built a modular SVM class that covers training, evaluation, and visualization.
- Creating visualizations of decision boundaries for linear, polynomial, and RBF kernels.
- Analyzed kernel performance with insights for tuning and multiclass scaling.
- **Tools Used:** Python, NumPy, Matplotlib, Scikit-learn.

Expenses Tracker Web App

[Github Repo](#)

- Developed a full-stack web application to manage income and expenses with JWT-based authentication.
- Implemented features for adding, editing, deleting, and categorizing transactions, with real-time visualization.
- Built a responsive and accessible UI ensuring smooth user experience across devices.
- Designed system architecture diagrams using Excalidraw and created UI/UX mockups in Figma.
- **Tools Used:** React, Next.js, Tailwind CSS, Node.js, Express.js, MongoDB, Chart.js, JWT, Figma, Excalidraw.

Client-Server Chatbot

[Github Repo](#)

- Developed a real-time chatbot using Python sockets and Flask, supporting multiple clients via multithreading.
- Integrated LLaMA 3 through Groq API with per-user chat history and a simple web interface.
- **Tools Used:** Python, Flask, Sockets, Multithreading, LLaMA 3, Groq API, HTML/CSS.

Portfolio Website

[Github Repo](#)

- Built a responsive portfolio website using HTML, JavaScript, and Tailwind CSS with optimized loading and smooth interactive UI.
- Easily deployed via Vercel, ensuring fast and reliable access to the live demo.
- Organized project structure with clear assets, icons, and documentation for maintainability and easy customization.
- **Tools Used:** HTML5, Tailwind CSS, JavaScript, AOS, Vercel, Git/GitHub.

Skills

Programming Languages: Python, C++, C#, JavaScript, TypeScript, SQL, HTML/CSS, Tailwind CSS

AI & Machine Learning: Scikit-learn, TensorFlow, PyTorch, OpenCV, NumPy, Pandas, CNNs, transfer learning, model evaluation, clustering, regression, NLP (LLaMA, OpenAI API, Groq)

Development & Deployment: React, Node.js, MongoDB, Flask, FastAPI, REST APIs, Docker, CI/CD, socket programming, client-server architecture

Soft Skills: Problem Solving, Critical Thinking, Adaptability, Creativity, Initiative, Time Management, Client Communication, Teamwork

Languages

Arabic: Native

English: Professional Proficiency